

hrbacek 1

Vincent Wang

May 2026

1 Sets

3.1. There exists only one set with no elements.

Proof: By Axiom of Existence, there exists a set Z with no elements. Let Z' be a set with no elements. Let X be a set, so $X \notin Z$. Then there is nothing to show. Similarly, $X \notin Z'$, so $Z = Z'$ by Axiom of Extensionality.

3.4. Let A be a set and P a property. Then there exists a unique set B such that $x \in B$ iff $x \in A$ and $P(x)$.

Proof: Let B, B' have the property and $x \in B$. Then $x \in A$ and $P(x)$, so $x \in B'$. Similarly, if $x \in B'$, then $x \in B$. By Extensionality, $B = B'$.

Claim: Let P be any property. Then $B = \{x \in \emptyset : P(x)\} = \emptyset$. Let $x \in B$. Then $x \in \emptyset$ and $P(x)$, the former being absurd. Thus, x has no elements and equals $\emptyset$.

Axiom of pair. Given two sets A, B , there exists a set C such that $x \in C$ iff $x = A$ or $x = B$.

Proof of uniqueness. Suppose C' is another such set. Let $x \in C'$. If $x = A$, then $x \in C$. If $x = B$, then $x \in C$. Similarly, if $x \in C$, then $x \in C'$, so $C = C'$ by Extensionality.

Claim: $\{\emptyset, \{\emptyset\}\} \neq \{\emptyset\}$.

Proof: Suppose $\{\emptyset\} \in \{\emptyset\}$, so $\{\emptyset\} = \emptyset$, but $\emptyset \in \{\emptyset\}$ while $\emptyset \notin \emptyset$.

Axiom of union. For any set S , there exists a set U such that $x \in U$ iff $x \in A$ for some $A \in S$.

Claim: Let A be a set and P a property such that $P(x) \rightarrow x \in A$. Then the set $\{x \in A : P(x)\}$ is independent of A .

Proof: Let A' be another such set. Suppose $x \in \{x \in A : P(x)\}$, so $P(x)$

and $x \in A$. Then $x \in A'$, so $x \in \{x \in A' : P(x)\}$. The other direction is similar, so by Extensionality $\{x \in A : P(x)\} = \{x \in A' : P(x)\}$.

Claim: $\{x : x \in P \text{ and } x \in Q\}$ exists.

Proof: Suppose $P(x)$, so $x \in P$.

E 3.1. The set of all $x \in A$ and $x \notin B$ exists.

Proof: Let $P(x, A, B)$ denote the property $x \in A$ and $x \notin B$. If $P(x, A, B)$ holds, then $x \in A$. Hence, this set exists.

E 3.2. Weak axiom of existence implies axiom of existence.

Proof: Let A be a set that exists. Let $A' = \{x \in A : x \neq x\}$. If $x \in A'$, then $x \neq x$, absurd. Hence, A' has no elements, so an empty set exists.

E 3.3. (a). Prove that a set of all sets does not exist.

Proof: Suppose V is a set such that for all x , $x \in V$. Let $W = \{x \in V : x \notin x\}$. Then $x \in W$ iff $x \notin x$. Take $x = W$, so $W \in W$ iff $W \notin W$, absurd.

For all sets A , there exists $x \notin A$.

E 3.5. $\{A, B, C\}$ exists.

Proof: $\{A, B\}, \{C\}$ exist by Pairing. Let $X = \{A, B\} \cup \{C\}$ by Union. Suppose $x \in X$. Then $x \in \{A, B\}$ or $x \in \{C\}$. For the first, $x = A$ or $x = B$. For the second, $x = C$.

Now if $x = A$ or $x = B$ or $x = C$, the first two give $x \in \{A, B\}$. The last gives $x \in \{C\}$, which means $x \in \{A, B\} \cup \{C\}$.

Claim: $\mathcal{P}(X) \subseteq X$ is false for all X .

Proof: Suppose $\mathcal{P}(X) \subseteq X$. Define $Y = \{x \in X : x \notin x\}$. We have $Y \in \mathcal{P}(X)$, so $Y \in X$. $x \in Y$ iff $x \notin x$. Take $x = Y$, so $Y \in Y$ iff $Y \notin Y$, absurd.

Claim: Weak forms of the axioms imply usual forms.

Proof: Axiom of Pair. Let C be a set such that $A \in C$ and $B \in C$. Let $C' = \{x \in C : x = A \text{ or } x = B\}$. If $x = A$ or $x = B$, then $x \in C'$. Conversely, if $x \in C$ and $x = A$ or $x = B$, then $x = A$ or $x = B$.

Axiom of Union. Let S be a set and U be a set such that $x \in A$ for $A \in S$ implies $x \in U$. Let $U' = \{x \in U : \text{there exists } A \in S \text{ such that } x \in A\}$.

Axiom of Powerset. Let $P' = \{x \in P : x \subseteq S\}$.

Claim: $\{\{a\}, \{a, b\}\} = \{\{a'\}, \{a', b'\}\}$ iff $a = a'$ and $b = b'$.

Proof: First, we show if $x \neq y$, then $\{x, y\} \neq \{z\}$. Suppose $\{x, y\} = \{z\}$. Then $x = z$ and $y = z$, so $x = y$, absurd. If $a = b$, then $\{a\} = \{a, b\}$, so $\{a'\} = \{a', b'\}$, so $a' = b'$. We have $\{\{a\}\} = \{\{a'\}\}$, so $\{a\} = \{a'\}$, so $a = a'$. Now suppose $a \neq b, a' \neq b'$. Then $\{a\} = \{a'\}$ or $\{a\} = \{a', b'\}$. Since $a' \neq b'$, we have $\{a\} = \{a'\}$, so $a = a'$. Similarly, $\{a, b\} = \{a', b'\}$. We have $b' = a$ or $b' = b$. If $b' = a$, then $\{a', b'\} = \{a\}$, absurd. Hence, $b' = b$. The converse is true.

Claim: $(a, b, c) = (a', b', c')$ iff $a = a', b = b', c = c'$.

Proof: One direction is clear. Suppose $(a, b, c) = (a', b', c')$. Then $(a, b) = (a', b')$ and $c = c'$. The first condition implies $a = a'$ and $b = b'$.

Claim: $(a, b) \in \mathcal{P}(\mathcal{P}(A))$ if $a, b \in A$.

Proof: $\{a\}, \{a, b\} \in \mathcal{P}(A)$, so $(a, b) = \{\{a\}, \{a, b\}\} \in \mathcal{P}(\mathcal{P}(A))$.

Cartesian product. Let X, Y be sets. Define $X \times Y = \{z \in \mathcal{P}(\mathcal{P}(X \cup Y)) : z = (x, y) \text{ for some } x \in X \text{ and } y \in Y\}$.

Let $R \subseteq X \times Y$. R is said to be a relation. Define

$$\begin{aligned} \text{dom}(R) &= \{x \in X : \text{there exists } y \in Y \text{ such that } (x, y) \in R\} \\ \text{rng}(R) &= \{y \in Y : \text{there exists } x \in X \text{ such that } (x, y) \in R\} \\ \text{field}(R) &= \text{dom}(R) \cup \text{rng}(R) \end{aligned}$$

Claim: Let $B \subseteq X$ and R a relation on X . Then the image of the inverse of R on B equals the inverse image of B .

Proof: Let T denote the inverse of R . We wish to show $T[B] = R^{-1}[B]$. Suppose $x \in T[B]$, so $(y, x) \in T$ for some $y \in B$. Then $(x, y) \in R$, so $x \in R^{-1}[B]$. Conversely, if $x \in R^{-1}[B]$, then $(x, y) \in R$ for some $x \in B$, so $(y, x) \in T$, so $x \in T[B]$, as desired.

Claim: Let R be a relation and $A = \bigcup(\bigcup R)$. Suppose $(x, y) \in R$. Then $x, y \in A$.

Proof: We have $\{\{x\}, \{x, y\}\} \subseteq \bigcup R$, so $\{x, y\} \in \bigcup R$, so $x, y \in \bigcup(\bigcup R) = A$. Hence, $\text{dom}R, \text{rng}R$ exist.